

Tolerance analysis. Nominal component values are specified in the designs. It is, however, practically impossible to manufacture all the components exactly with required values. Hence allowable tolerances (upper and lower limits) are to be specified for the component values so that even when actual component values vary within designated tolerance limits, the design performs as required.

Simulate the design with component values set to extreme limits within the specified tolerance. Confirm that the design still performs well. Correct any non-conformance by modifying the design and simulating again.

Monte Carlo analysis. During production, all components in a particular unit will not be at their extreme tolerance limit values. In fact, actual values will vary from unit to unit randomly within the tolerance limit. This random variation can be modeled by Monte Carlo analysis.

Monte Carlo analysis simulates a hundred or thousand runs of the same schematic. It varies component values randomly within the set tolerance limits as per set distribution from schematic to schematic. The results can be analysed to determine the percentage of circuits that meet specification limits.

The results help determine whether the selected tolerance for components is adequate or not and whether it can be widened for some components. Wider-tolerance components cost less than narrow-tolerance components.

Margin (de-rating) analysis. A designed unit is required to function during its operational lifetime under expected operational stresses. The capability to withstand stresses (temperature, vibration, etc) varies from component to component. Component values and capabilities also vary/degrade with storage and time because of chemical and physical phenomena.

To accommodate these variations, units are to be designed to withstand more stress than required during operation. For example, if a unit is required to operate at 50°C, components that

Quality and reliability defined in different ways

Quality

Definition 1: Conformance to requirements—given by Crosby

Definition 2: Freedom from deficiencies or fitness for use—given by Juran

Definition 3: Degree to which a set of inherent characteristics of a product fulfils requirements—given by International Standards Organization

Reliability

The probability that an item performs its 'designed function' for a 'stated period of time,' when operated under 'stated conditions.' It is a number between 0 and 1.

Let us try to understand statistics-based probabilistic definition of reliability more clearly. Say, a manufacturer sold a thousand refrigerators in a day all over the country. All started functioning on the same day. All were used as per the prescribed operating procedure in the prescribed environment for three years. Let us assume that 30 refrigerators in the batch failed in three years. It means that the probability of survival for three years is $(1000-30)/1000=970/1000=0.97$.

Hence 0.97 is the reliability for three-year time period for these refrigerators. The higher the reliability, lesser the probability of failure.

can withstand 70°C should be selected. Similarly, if a resistor consumes 0.25 watt during operation, a 0.5-watt resistor should be selected. This method of using components at less capability during operation is called de-rating. This is one way to design a reliable product.

One should also ascertain by simulation that the design performs satisfactorily at higher thermal (temperature) and dynamic (vibration) stresses than required during operation. This can be achieved by performing thermal and packaging analysis using EDA tools. This analysis is also called Margin analysis, as it ascertains that the design has margin compared to required operational stresses.

Printed circuit board layout and analysis. PCB layout pattern and design play an important role in the reliable performance of an electronic system. This is particularly true at high frequencies, where PCB tracks behave like transmission lines. This condition happens if signal transmission delay along the length of the PCB track is comparable with the rise and fall times of pulses. Pulses at the receiver will be distorted with ringing and not received by the receiver correctly. Pre-layout and post-layout signal integrity (SI) analysis using EDA tools needs to be done to examine this problem. Terminating resistors, if required, need to be added at appropriate places.

Care should be taken so that the

loop area enclosed by signal current's send and return paths is minimised. Larger loop area results in electromagnetic interference (EMI) to nearby circuits through mutual inductive coupling. It also makes the circuit under operation susceptible to EMI from nearby circuits.

Signal return current path is normally the circuit's signal reference ground. This ground is shared by many signals. Ideally, ground should have zero impedance at all frequencies. Practically, ground tracks will have non-zero impedance (inductive), especially at higher frequencies. This results in non-zero voltage drop across ground, causing the transmitted signal to be received erroneously by the receiver. Care should be taken to ensure near-zero-impedance ground at direct currents to frequencies of interest. To achieve this, a large ground area should be used. If necessary, use a multilayer PCB with separate planes for ground and power supply.

Keep the following things in mind:

1. Use separate grounds for sensitive analogue circuits and noisy digital circuits.
2. Physically separate sensitive analogue circuits from noisy digital circuits.
3. The rise and fall times of pulses should not be less than required. Pulses with short rise and fall times will have a high frequency content and